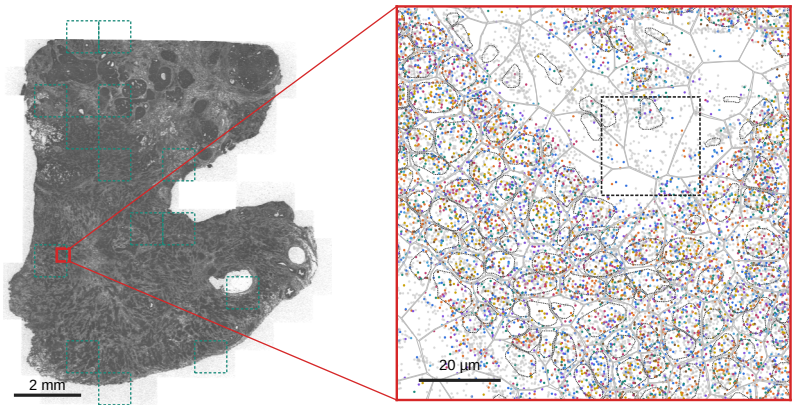
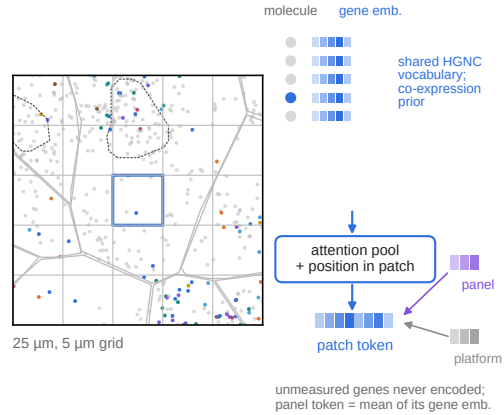
a **Molecules from one Xenium section, no segmentation**



68.1 M molecules · 280 genes
dashed: held-out 1 mm blocks (eval only)

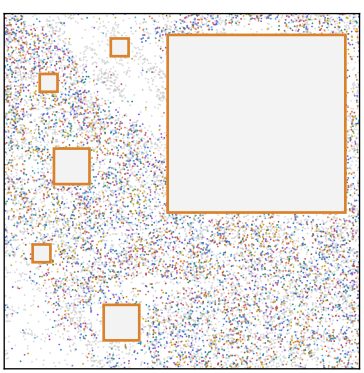
100 µm window; colour = gene (top 8)
outlines: 10x cells / nuclei, eval only

b **5 µm patches become tokens on any panel**

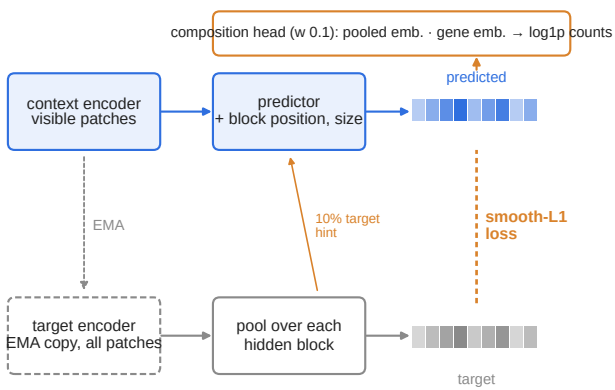


unmeasured genes never encoded;
panel token = mean of its gene emb.

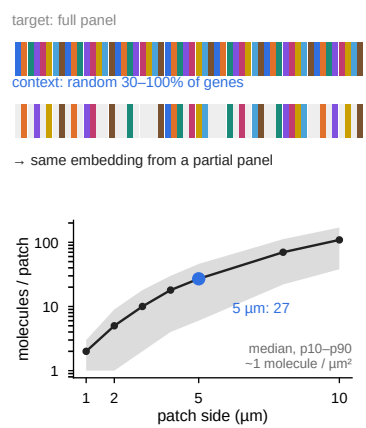
c **Multiscale JEPA: predict hidden regions' embeddings from context**



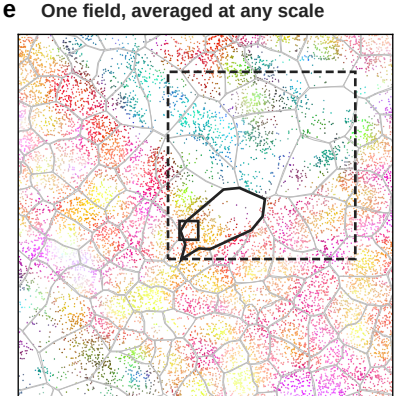
hidden blocks of 5, 10 and 50 µm
(about 60% of the window)



d **Panel dropout and token size**



e **One field, averaged at any scale**



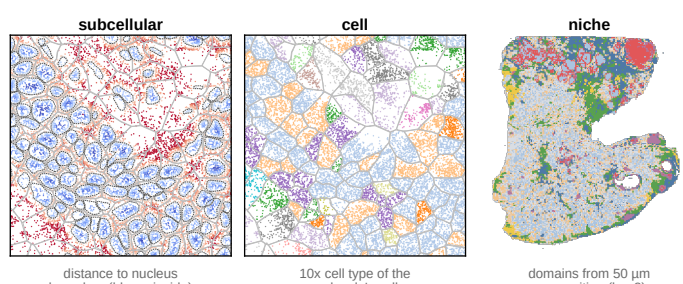
5 µm spot
mean of E(x) →
subcellular

cell (any outline)
mean of E(x) →
cell

50 µm window
mean of E(x) →
niche

schematic of E(x): colours from local composition
(stand-in; the trained field is not shown)

f **Linear probes per scale; the bar is composition at that scale**



Bar to beat, measured with comp20 (~2.6 µm composition):
nucleus AUROC 0.82 · nucleus-distance ρ 0.62 · cell-distance ρ 0.33